

Fraud Detection in Credit Card Transactions

Introduction

Credit card fraud is a significant challenge for financial institutions, with fraudulent transactions making up a very small fraction of overall transaction volume. This project explores how machine learning techniques can be applied to automatically identify fraudulent transactions from a large dataset of anonymized credit card transactions, using both unsupervised anomaly detection and supervised classification approaches.

Abstract

The objective of this project was to build a fraud detection system capable of distinguishing normal transactions from fraudulent ones despite extreme class imbalance in the dataset. Two anomaly detection methods, Isolation Forest and Local Outlier Factor, were applied to detect fraud without relying on labeled data, while a supervised XGBoost classifier was trained to learn directly from labeled examples. Model performance was evaluated using precision, recall, F1-score, and ROC-AUC rather than accuracy, since accuracy is a misleading metric on highly imbalanced data. The final trained model was deployed through an interactive Streamlit dashboard that allows transaction CSV files to be uploaded and classified in real time.

Tools Used

- Python – core programming language
- Pandas & NumPy – data loading, cleaning, and preprocessing
- Scikit-learn – Isolation Forest, Local Outlier Factor, StandardScaler, evaluation metrics
- XGBoost – supervised gradient boosting classifier
- Streamlit – interactive web dashboard for live predictions
- Joblib – saving and loading the trained model

Steps Involved in Building the Project

1. Loaded the Credit Card Fraud Detection dataset (Kaggle, mlg-ulb) containing 284,807 transactions with 31 features, and confirmed extreme class imbalance (approximately 0.17% fraud cases).
2. Preprocessed the data by scaling the Amount and Time features using StandardScaler, since the remaining features (V1–V28) were already PCA-transformed and scaled.
3. Split the dataset into training and test sets using stratified sampling to preserve the original fraud ratio in both sets.
4. Applied Isolation Forest and Local Outlier Factor as unsupervised anomaly detection baselines to identify fraudulent transactions without using labels.
5. Trained a supervised XGBoost classifier on the labeled training data.
6. Evaluated all models using precision, recall, F1-score, and ROC-AUC, with particular focus on recall for the fraud class, since missed fraud cases are more costly than false alarms.
7. Saved the trained XGBoost model using Joblib and built a Streamlit dashboard that allows a transaction CSV file to be uploaded, automatically preprocessed, and classified as

fraudulent or normal in real time.

Conclusion

This project demonstrated that supervised learning with XGBoost significantly outperforms unsupervised anomaly detection methods when labeled fraud data is available, since it can directly learn the distinguishing patterns of fraudulent transactions. It also highlighted the importance of using appropriate evaluation metrics, such as recall and ROC-AUC, when working with highly imbalanced datasets, as accuracy alone would be misleading. The resulting Streamlit dashboard shows how such a model could be integrated into a practical, interactive tool for real-time transaction screening.